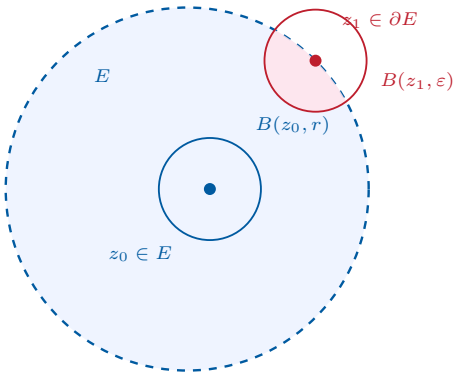
复变函数答案

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1　习题一

16. 证明:



粉色区域为 $B(z_1,\varepsilon) \cap E \neq \varnothing$;
而 $B(z_0,r) \subseteq E$

一、$\partial E \cup E \subseteq E'$

1° $E \subseteq E'$:
设 $z_0 \in E$.
$\because E$ 为开集,
$\therefore \exists r > 0$ 使 $B(z_0, r) \subseteq E$.
$\forall \varepsilon > 0$, 取 $w = z_0 + \frac{1}{2}\min(r, \varepsilon)$,
则 $0 < |w - z_0| < \min(r, \varepsilon)$,
$\therefore w \in E$ 且 $w \in B(z_0, \varepsilon) \setminus \{z_0\}$,
$\therefore z_0 \in E'$, 即 $E \subseteq E'$.

2° $\partial E \subseteq E'$:
设 $z_0 \in \partial E$.
若 $z_0 \in E$, 由 1° 知 $z_0 \in E'$;
若 $z_0 \notin E$,
$\because z_0 \in \partial E$,
$\therefore \forall \varepsilon > 0, B(z_0, \varepsilon) \cap E \neq \varnothing$,
且其中各点均异于 z_0,
$\therefore \big(B(z_0, \varepsilon) \setminus \{z_0\}\big) \cap E \neq \varnothing$,
$\therefore z_0 \in E'$.
综上, $\partial E \cup E \subseteq E'$.

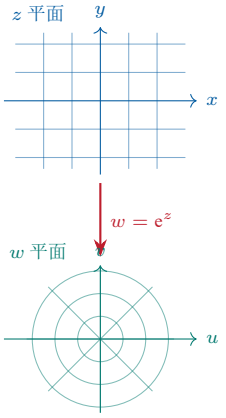
二、$E' \subseteq \partial E \cup E$

设 $z_0 \in E'$.
若 $z_0 \in E$, 结论已成立;
若 $z_0 \notin E$,
$\because z_0 \in E'$,
$\therefore \forall \varepsilon > 0, \big(B(z_0, \varepsilon) \setminus \{z_0\}\big) \cap E \neq \varnothing$,
$\therefore B(z_0, \varepsilon) \cap E \neq \varnothing$;
又 $\because z_0 \notin E$,
$\therefore z_0 \in B(z_0, \varepsilon) \cap E^c$,
$\therefore B(z_0, \varepsilon) \cap E^c \neq \varnothing$,
$\therefore z_0 \in \partial E$.
综上, $E' \subseteq \partial E \cup E$.

综合一、二, $E' = \partial E \cup E$. ■

4. 解: 设 $z = x + iy$, $f(z) = u(x,y) + iv(x,y)$, Cauchy-Riemann 条件为

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}, \quad \text{即} \quad u_x = v_y,\ u_y = -v_x.$$



以 (4) 为例: 水平线、竖直线分别映为射线、圆, 且正交相交 (C-R 条件的几何意义)

(1) $\sin z$:

$\because \sin z = \sin x \cosh y + i \cos x \sinh y$,

$\therefore u = \sin x \cosh y, \quad v = \cos x \sinh y$.

$\because u_x = \cos x \cosh y = v_y$,

$u_y = \sin x \sinh y = -v_x$,

$\therefore$ 满足 C-R 条件.

(2) $\cos z$:

$\because \cos z = \cos x \cosh y - i \sin x \sinh y$,

$\therefore u = \cos x \cosh y, \quad v = -\sin x \sinh y$.

$\because u_x = -\sin x \cosh y = v_y$,

$u_y = \cos x \sinh y = -v_x$,

$\therefore$ 满足 C-R 条件.

(3) z^3:

$\because z^3 = (x^3 - 3xy^2) + i(3x^2y - y^3)$,

$\therefore u = x^3 - 3xy^2, \quad v = 3x^2y - y^3$.

$\because u_x = 3x^2 - 3y^2 = v_y$,

$u_y = -6xy = -v_x$,

$\therefore$ 满足 C-R 条件.

(4) e^z:

$\because \mathrm{e}^z = \mathrm{e}^x \cos y + i\, \mathrm{e}^x \sin y$,

$\therefore u = \mathrm{e}^x \cos y, \quad v = \mathrm{e}^x \sin y$.

$\because u_x = \mathrm{e}^x \cos y = v_y$,

$u_y = -\mathrm{e}^x \sin y = -v_x$,

$\therefore$ 满足 C-R 条件.

综上, 四个函数的实部与虚部在 $\mathbb{C}$ 上处处满足 C-R 条件.

5. 证明:

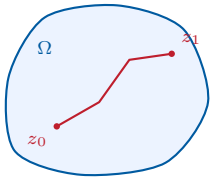
引理（常值判别）: 若实值函数 u 在区域 Ω 内恒有 $u_x = u_y = 0$, 则 u 在 Ω 内为常数.

$\because \Omega$ 为连通开集,

$\therefore \forall z_0, z_1 \in \Omega$, 可用完全含于 Ω 的折线联结.

每段折线上以 t 为参数, 则 $\dfrac{\mathrm{d}u}{\mathrm{d}t} = u_x\, x'(t) + u_y\, y'(t) = 0$,

$\therefore u(z_0) = u(z_1)$, 即 u 为常数.



区域连通: 任两点可用含于 Ω 的折线联结

(1) 设 $f = u + \mathrm{i}v$.

$\because f' = u_x + \mathrm{i}v_x = 0$,

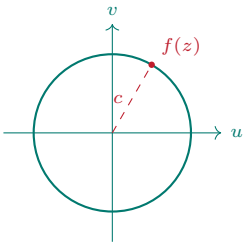
$\therefore u_x = v_x = 0$;

又 $\because f$ 解析, $\therefore$ 满足 C-R 条件: $u_y = -v_x$, $v_y = u_x$,

$\therefore u_y = v_y = 0$.

$\therefore u, v$ 的一阶偏导在 Ω 内恒为零,

由引理, u, v 均为常数, $\therefore f$ 在 Ω 内为常数.



(2): 像点恒落在圆周 $|w| = c$ 上

(2) 设 $f = u + \mathrm{i}v$, 则 $|f|^2 = u^2 + v^2 = c^2$ ($c \geq 0$ 为常数).

若 $c = 0$, 则 $f \equiv 0$, 为常数.

下设 $c > 0$. 两边分别对 x, y 求偏导:

$\because uu_x + vv_x = 0$,

$uu_y + vv_y = 0$,

将 C-R 条件 $u_y = -v_x$, $v_y = u_x$ 代入第二式, 得 $vu_x - uv_x = 0$.

$\therefore$

$$\begin{pmatrix} u & v \\ v & -u \end{pmatrix} \begin{pmatrix} u_x \\ v_x \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

$\because$ 系数行列式 $= -(u^2 + v^2) = -c^2 \neq 0$,

$\therefore u_x = v_x = 0$, 进而 $u_y = v_y = 0$.

由引理, u, v 均为常数, $\therefore f$ 在 Ω 内为常数. ■